

HOMEWORK 5

1. Let $R \rightarrow S$ be a ring homomorphism. Let M be a free (left) R -module with basis X . Prove that the map $M \rightarrow S \otimes_R M$ taking m to $1 \otimes m$ restricts to a bijection between X and $1 \otimes X$ and the (left) S -module $S \otimes_R M$ is free with basis $1 \otimes X$.
2. In the situation $(M_R, {}_R N)$ consider the R -module homomorphism $f : R^{(M)} \rightarrow M$ taking a basis element m from M to $m \in M$ and a similar R -module homomorphism $g : R^{(N)} \rightarrow N$. Prove that the tensor product of f and g is surjective. Derive the fact that the abelian group $M \otimes_R N$ is generated by the elements $m \otimes n$.
3. In the situation $(M_R, {}_R N_S, {}_S P)$ prove that there is a natural group isomorphism $(M \otimes_R N) \otimes_S P \simeq M \otimes_R (N \otimes_S P)$.
4. Let S be a multiplicative subset of a commutative ring R . Prove that for an R -module M the $S^{-1}R$ -modules $S^{-1}M$ and $M \otimes_R S^{-1}R$ are naturally isomorphic.
5. Prove that if every module over a domain R is free, then R is a field.
6. Prove that the ideal I in $R = \mathbb{Z}[\sqrt{-5}]$ generated by 2 and $1 + \sqrt{-5}$ is a projective R -module. Is I a free R -module?
7. Let $f : A \rightarrow A$ be an endomorphism of an abelian group A such that $f(f(a)) = -a$ for all $a \in A$. Show that there is structure of a $\mathbb{Z}[i]$ -module on A such that $ia = f(a)$ for all $a \in A$.
8. Let N be a submodule of a free finitely generated module F over a PID R . Show that N is a direct summand of F if and only if $N \cap aF = aN$ for all $a \in R$.
9. A right R -module M is called *flat* if the functor $R\text{-Mod} \rightarrow Ab$ taking a left R -module N to $M \otimes_R N$ is exact.
 - a) Prove that projective modules are flat.
 - b) Prove that $\mathbb{Q}$ is a flat abelian group but not projective.
10. Let S be a multiplicative subset of a commutative ring R . Prove that the R -module $S^{-1}R$ is flat.